

Program : Diploma in Cloud Computing and Big Data	
Course Code : 6278	Course Title: Advanced Cloud Computing Lab
Semester : 6	Credits: 1.5
Course Category: Programme Core Course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To impart the knowledge of CloudSim to learn cloud computing architecture/system concepts
- To make use of public cloud platforms providing IaaS, PaaS and SaaS services

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics of Cloud Computing Concepts	3273	Introduction to Cloud Computing	3
Virtual Machine Launching	3276	Cloud Computing Lab	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate basic structure and syntax of Java and implement simulation of different cloud computing architecture/system concepts	9	Applying
CO2	Implement Simulation of Data Centers and Network Topologies on CloudSim	8	Applying
CO3	Demonstrate launching of IaaS cloud	8	Applying

CO4	Develop applications on PaaS cloud and SaaS cloud	14	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			3
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate basic structure and syntax of Java and implement simulation of different cloud computing architecture/system concepts		
M1.01	Configure CloudSim	2	Applying
M1.02	Demonstrate basic structure and syntax of Java Programming Language	2	Applying
M1.03	Execute Basic Examples on CloudSim	5	Applying

Contents:

- Download and Install CloudSim.
- Basic structure and Syntax of Java to understand the CloudSim code.
- A simple example showing how to create a datacenter with one host and run one cloudlet on it.
- A simple example showing how to create two datacenters with one host and a network topology each and run two cloudlets on them.
- A simple example showing how to create two datacenters with one host each and run cloudlets of two users with network topology on them.
- A simple example showing how to create two datacenters with one host each and run two cloudlets on them.
- A simple example showing how to create two datacenters with one host each and run cloudlets of two users on them.

CO2	Implement Simulation of Data Centers and Network Topologies on CloudSim		
M2.01	Execute Network Examples on CloudSim	8	Applying

Contents:

- A simple example showing how to create a datacenter with one host and a network topology and and run one cloudlet on it.
- A simple example showing how to create two datacenters with one host and a network topology each and run two cloudlets on them.
- A simple example showing how to create two datacenters with one host each and run cloudlets of two users with network topology on them.
- A simple example showing how to create a datacenter with one host and a network topology and and run one cloudlet on it. Here, instead of using a BRIE file describing the links, links are inserted in the code.

CO3	Demonstrate launching of IaaS cloud		
M3.01	Configure an AWS account.	2	Applying
M3.02	Use the AWS Management Console to launch an EC2 instance	3	Applying
M3.03	Experiment for making Connection to EC2 Instance	3	Applying
	Lab Test – I	3	Applying

CO4	Develop applications on PaaS cloud and SaaS cloud		
M4.01	Develop simple applications using Google App Engine	4	Applying
M4.02	Practice simple Windows Azure Applications	4	Applying
M4.03	Collaborate on Google Docs	2	Applying
M4.04	Collaborate on Google Sheets	2	Applying
M4.05	Demonstrate the use of DropBox and Google Drive	2	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
R1	Mark Wilkins, Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud, Addison-Wesley Professional; 1st edition, 2019
R2	Andreas Wittig, Michael Wittig, Amazon Web Services in Action, Manning; 2nd edition, 2018
R3	Dan Sanderson, Programming Google App Engine, O'Reilly Media, Inc., 2009

Online Resources

Sl.No	Website Link
1	https://www.geeksforgeeks.org/what-is-cloudsim/
2	http://www.cloudbus.org/cloudsim/
3	https://aws.amazon.com/
4	https://cloud.google.com/appengine/docs/flexible/go/create-app
5	https://azure.microsoft.com/en-us/solutions/low-code-application-development/